

HW 3

$$\textcircled{1} \quad a[n] = \begin{matrix} & -3 & -2 & -1 & 0 \\ & -5 & 3 & -1 & 2 \\ & & & \uparrow & \end{matrix}$$

$$b[n] = \begin{matrix} & 1 & 2 & 3 & 4 \\ -1 & 2 & -2 & 2 \\ & & & \uparrow & \end{matrix} \Rightarrow b[-n] = \begin{matrix} & -4 & -3 & -2 & -1 \\ 2 & -2 & 2 & -1 \\ & & & \uparrow & \end{matrix}$$

$$y[n] = \begin{matrix} & 1 & 2 & 3 & 4 \\ 5 & -13 & 17 & -20 & 12 & -6 & 4 \\ & & & \uparrow & \end{matrix} \quad -2 \leq n \leq 4$$

$$\textcircled{2} \quad h[n] \otimes x[n] = y[n] \quad y[n] = \begin{matrix} & 2 & & & & & & 9 \\ 6 & 11 & -13 & 16 & 1 & 9 & 2 & 8 \end{matrix} \quad 2 \leq n \leq 9$$

$$h[n] = \begin{matrix} & 1 & & & 4 \\ 2 & 5 & -1 & 4 \end{matrix} \quad 1 \leq n \leq 4$$

$$h[z] = 4z^{-4} - z^{-3} + 5z^{-2} + 2z^{-1}$$

$$x[z] = \frac{y[z]}{h[z]} \cdot \frac{z^9}{z^9} = \frac{6z^7 + 11z^6 - 13z^5 + 16z^4 + z^3 + 9z^2 + 2z + 8}{2z^8 + 5z^7 - z^6 + 4z^5}$$

$$\begin{array}{r} 2z^8 + 5z^7 - z^6 + 4z^5 \overline{) 6z^7 + 11z^6 - 13z^5 + 16z^4 + z^3 + 9z^2 + 2z + 8} \\ \underline{-6z^7 - 15z^6 + 3z^5 - 12z^4} \\ 0 - 4z^6 - 10z^5 + 4z^4 + z^3 \\ \underline{+ 4z^6 + 10z^5 - 2z^4 + 8z^3} \\ 0 + 0 + 2z^4 + 9z^3 + 9z^2 + 2z \\ \underline{- 2z^4 - 5z^3 + z^2 - 4z} \\ 0 + 4z^3 + 10z^2 - 2z + 8 \\ \underline{- 4z^3 - 10z^2 + 2z - 8} \\ 0 \end{array}$$

$$x[n] = [3, -2, 0, 1, 2] \quad 1 \leq n \leq 5$$

$$\textcircled{3} \quad y[n] = x[n] - 2x[n-3] + 0.5x[n-5] + 0.7x[n-7]$$

$$h[n] = \delta[n] - 2\delta[n-3] + 0.5\delta[n-5] + 0.7\delta[n-7]$$

$$\text{or } [1, 0, 0, -2, 0, 0.5, 0, 0.7]$$